GMI German-Malaysian Institute

DIPLOMA PROGRAMME

PRACTICAL TEST 2

Academic Period : July 2025

Subject	:	**OPERATING SYSTEMS ADMINISTRATION**
Code	:	**CIT 0633**
Duration	:	**4 HOURS**
Total Marks	:	**100 MARKS**

GENERAL INSTRUCTIONS:

1. Write down clearly (with capital letters) your name, trade, group and the date of examination at the bottom of this page.
2. Do not open this question booklet until you are told to do so by the invigilator.
3. Read the specific instructions and questions carefully before answering.

Student's Name	:	**ABDUL HARIZ BIN MOHD SARNUDIN**
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Semester/Trade/Group	:	**SEMESTER 3 / DCBS / DCBS 1**
Date	:	**23/10/2025**

No	Assessment Criteria	CLO	PO	Level of Diff	Full Marks	Score	Marks	Comment
1.	Installation of Centos Server	2	2	P3	10			
2	i. Configuring DNS Server ii. Configuring Web Server iii. Configuring DHCP Server	2	3, 10	P3	75			
3.	Troubleshooting	2	4	P3	10			
4.	Knowledge	2		P3	5			
				Total	100			
				Total Marks	/100			
				Signature				

Learning Objectives

- To do installation of Centos 10 Stream Server
- To configure DNS Server
- To configure Web Server
- To configure DHCP Server

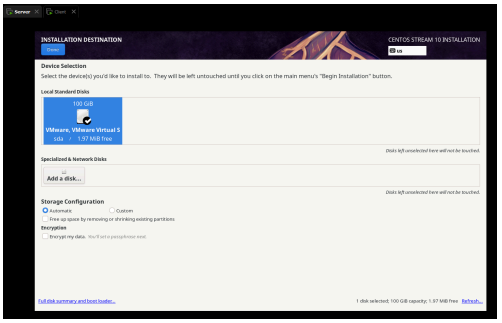
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Rubrics

NO.	CRITERIA	APPROACHES EXPECTATIONS [1]	MEETS EXPECTATIONS [2]	EXCELLENT [3]	MARKS
PRACTICAL TEST (2)					
1	**INSTALL LINUX SERVER**	Unable to install Linux server	Able to install Linux server	Able to install Linux server and successful ping to another Linux client	
2	**CONFIGURE DNS SERVER**	Unable to configure DNS server	Able to to configure DNS server only	Able to configure DNS server and client	
3	**CONFIGURE WEB SERVER**	Unable to configure Web server	Able to configure Web server only	Able to configure Web server and test from Linux client	
4	**CONFIGURE DHCP SERVER**	Unable to configure DHCP server	Able to configure DHCP server only	Able to configure DHCP server and test from Linux client	
5	**KNOWLEDGE**	Only restatement of the instruction without commenting on the expected key point. Incorrect judgment/arguments are used.	Shows a little understanding of what the practical was about.	Shows a full understanding of what the practical was about.	
	TOTAL				

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Tasks	Evidences (Please screenshot the result as evidence)	Marks

Install Linux Server
100GB Disk, 4GB RAM
 	
Hostname: h4rz-server
IP Server: 172.30.1.110
Hostname: h4rz-client
IP Server: 172.30.1.213
Domain: cerberix.my
	Server Installation :
	 
	 
	
	Server IP and Domain :
	`inet 172.30.1.110/16`
	`valid_lft forever`
	`inet6 fe80::20c:29ff:`
	`valid_lft forever`
	`root@h4rz-server:~#`
	
	`root@h4rz-server:~# hostname -d`
	`mycentos.net`
	`root@h4rz-server:~#`
	
	Client IP and Domain :
	`inet 172.30.1.213/24`
	`33`
	`valid_lft 5777sec`
	`inet6 fe80::20c:29ff:`
	`valid_lft forever`
	`root@h4rz-client:~#`
	
	`root@h4rz-client:~# hostname -d`
	`mycentos.net`
	`root@h4rz-client:~#`
		10

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Able to configure DNS Server
• Edit main configuration file /etc/named.conf
• Set forward and reverse lookup zones
• Name resolution can be done in Client
	Install DNS service (bind) :
	`root@h4rz-server:~# dnf install bind bind-utils -y`
	`Last metadata expiration check: 2:37:22 ago on Thu 23 Oct 2025 09:19:58 AM +08.`
	`Package bind-32:9.18.33-5.el10.x86_64 is already installed.`
	`Package bind-utils-32:9.18.33-5.el10.x86_64 is already installed.`
	`Dependencies resolved.`
	`Nothing to do.`
	`Complete!`
	`root@h4rz-server:~#`
	
	DNS (named.conf) configuration :
	`root@h4rz-server:~# cat /etc/named.conf`
	`//`
	`// named.conf`
	`//`
	`// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS`
	`// server as a caching only nameserver (as a localhost DNS resolver only).`
	`//`
	`// See /usr/share/doc/bind*/sample/ for example named configuration files.`
	`//`
	
	`options {`
	`listen-on port 53 { 127.0.0.1; 172.30.1.110; };`
	`listen-on-v6 port 53 { ::1; };`
	`directory "/var/named";`
	`dump-file "/var/named/data/cache_dump.db";`
	`statistics-file "/var/named/data/named_stats.txt";`
	`memstatistics-file "/var/named/data/named_mem_stats.txt";`
	`secroots-file "/var/named/data/named.secroots";`
	`recursing-file "/var/named/data/named.recursing";`
	`allow-query { localhost; 172.30.1.0/24; };`
	
	`/*`
	
	`// Forward Zone - mycentos.net`
	`zone "centos.net" IN {`
	`type master;`
	`file "mycentos.net.zone";`
	`allow-update { none; };`
	`};`
	
	`/ Reverse Zone - 172.30.1.x`
	`zone "1.30.172.in-addr.arpa" IN {`
	`type master;`
	`file "1.30.172.rev.zone";`
	`allow-update { none; };`
	`};`
	`root@h4rz-server:~#`
	
	Set forward and reverse lookup zones :
	`root@h4rz-server:~# cat /var/named/mycentos.my.zone`
	`$TTL 86400`
	`@ IN SOA h4rz-server.mycentos.net. root.mycentos.net.(`
	`1 ; Serial`
	`3600 ; Refresh`
	`1800 ; Retry`
	`604800 ; Expire`
	`86400 ) ; Minimum TTL`
	
	`; Name Server`
	`@ IN NS h4rz-server.mycentos.net.`
	
	`; A Records (Name to IP)`
	`h4rz-server IN A 172.30.1.110`
	`h4rz-client IN A 172.30.1.215`
		10
		
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```
root@h4rz-server:~# cat /var/named/1.30.172.rev.zone
$TTL 86400
@       IN      SOA     h4rz-server.mycentos.net. root.mycentos.net. (
                1       ; Serial
                3600    ; Refresh
                1800    ; Retry
                604800  ; Expire
                86400 ) ; Minimul TTL

; Name Server
@       IN      NS      h4rz-server.mycentos.net.

; PTR Records (IP to Name)
110     IN      PTR     h4rz-server.mycentos.net.
215     IN      PTR     h4rz-client.mycentos.net.
```

Change directory ownership to named (dns service's system user) : (Alt : *chown named:named /var/named/mycentos.net.zone || chown named:named /var/named/100.30.172.rev.zone*)

```
root@h4rz-server:/var/www/html# chown named:named /var/named/mycentos.net.zone || chown named:named /var/named/10
.172.rev.zone
root@h4rz-server:/var/www/html#
```

Named resolution using nslookup on Client :

```
root@h4rz-client:~# nslookup h4rz-server.mycentos.net
Server:         172.30.1.110
Address:        172.30.1.110#53

Name:   h4rz-server.mycentos.net
Address: 172.30.1.110

root@h4rz-client:~# nslookup h4rz-client.mycentos.net
Server:         172.30.1.110
Address:        172.30.1.110#53

Name:   h4rz-client.mycentos.net
Address: 172.30.1.213

root@h4rz-client:~# nslookup 172.30.1.213
213.1.30.172.in-addr.arpa       name = h4rz-client.mycentos.net.

root@h4rz-client:~# nslookup 172.30.1.110
110.1.30.172.in-addr.arpa       name = h4rz-server.mycentos.net.

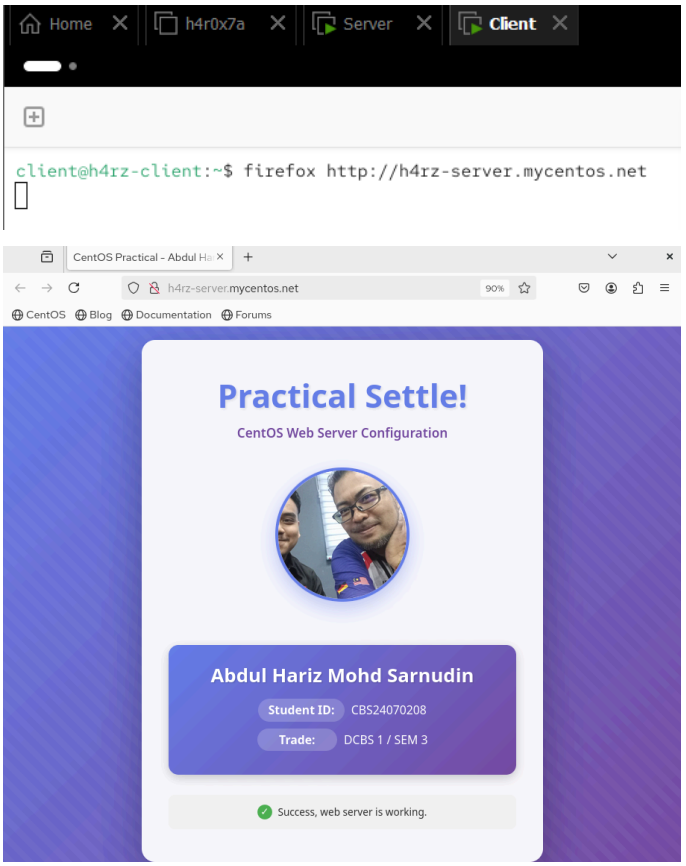
root@h4rz-client:~#
```


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Web Server
• Configure web server and DNS server
• Write simple website script
• Website can be viewed in client
	Install web server (httpd) :
	`root@h4rz-server:~# dnf install -y httpd`
	`Last metadata expiration check: 2:38:04 ago on Thu 23 Oct 2025 09:19:58 AM +08.`
	`Package httpd-2.4.63-5.el10.x86_64 is already installed.`
	`Dependencies resolved.`
	`Nothing to do.`
	`Complete!`
	`root@h4rz-server:~#`
	
	Configure virtual host :
	`root@h4rz-server:/var/www/main# cat /etc/httpd/conf.d/h4rz-server.mycentos.net.conf`
	`<VirtualHost *:80>`
	`ServerName h4rz-server.mycentos.net`
	`DocumentRoot /var/www/main`
	
	`<Directory /var/www/main>`
	`AllowOverride None`
	`Require all granted`
	`</Directory>`
	
	`ErrorLog /var/log/httpd/cbs1_error.log`
	`CustomLog /var/log/httpd/cbs1_access.log combined`
	`</VirtualHost>`
	`root@h4rz-server:/var/www/main#`
	
	Create directory and web page content :
	`root@h4rz-server:~# mkdir -p /var/www/main`
	`root@h4rz-server:~# cat /var/www/main/index.html`
	`<!DOCTYPE html>`
	`<html lang="en">`
	`<head>`
	`<meta charset="UTF-8">`
	`<meta name="viewport" content="width=device-width,`
	`<title>CentOS Practical - Abdul Hariz</title>`
	`<style>`
	`* {`
	`margin: 0;`
	`padding: 0;`
	`box-sizing: border-box;`
	`}`
	
	Change directory ownership, allow firewall and start httpd service :
	`root@h4rz-server:~# chown -R apache:apache /var/www/main`
	`root@h4rz-server:~# firewall-cmd --add-service=http --permanent`
	`Warning: ALREADY_ENABLED: http`
	`success`
	`root@h4rz-server:~# firewall-cmd --reload`
	`success`
	`root@h4rz-server:~# systemctl enable --now httpd`
	`root@h4rz-server:~#`
		15
		
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		5

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View on Client side :



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DHCP Server

- Configure DHCP server
- Client gets the IP address from server

Reminder: Please turn off WiFi access as not to disrupt GMI network

Install dhcp server (kea) :

```
root@h4rz-server:~# dnf install -y kea
Last metadata expiration check: 2:38:58 ago on Thu 23 Oct 2025 09:19:58 AM +08.
Package kea-3.0.0-2.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
root@h4rz-server:~#
```

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DHCP Configuration :

```
root@h4rz-server:~# cat /etc/kea/kea-dhcp4.conf
{
"Dhcp4": {
    // Listen on the server's interface
    "interfaces-config": {
        "interfaces": [ "ens33" ]
    },

    "control-socket": {
        "socket-type": "unix",
        "socket-name": "/run/kea/kea-dhcp4-ctrl.sock"
    },

    "lease-database": {
        "type": "memfile",
        "lfc-interval": 3600
    },

    // Define the subnet
    "subnet4": [
        {
            "subnet": "172.30.1.0/24",
            "id": 1,

            "pools": [
                { "pool": "172.30.1.210 - 172.30.1.220" }
            ],


   // Add reservations for 210, 211, 212.
   // Without a MAC address, they can't be assigned,
   // but this fulfills the "reservation" requirement.
   // These IPs will NOT be given to random clients.
   "reservations": [
       {
           "ip-address": "172.30.1.210",
           "hw-address": "00:11:22:33:44:55" // <-- Change to a real MAC
       },
       {
           "ip-address": "172.30.1.211",
           "hw-address": "00:11:22:33:44:66" // <-- Change to a real MAC
       },
       {
           "ip-address": "172.30.1.212",
           "hw-address": "00:11:22:33:44:77" // <-- Change to a real MAC
       }
   ],

   // Options to send to clients
```


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```
            "option-data": [
                {
                    "name": "routers",
                    "data": "172.30.1.1" // Gateway
                },
                {
                    "name": "domain-name-servers",
                    "data": "172.30.1.110" // Our DNS server
                },
                {
                    "name": "domain-name",
                    "data": "mycentos.net"
                }
            ]
        }
    ],

    "loggers": [
        {
            "name": "kea-dhcp4",
            "output_options": [
                {
                    "output": "/var/log/kea/kea-dhcp4.log"
                }
            ],
            "severity": "INFO"
        }
    ]
}
}
root@h4rz-server:~#
```

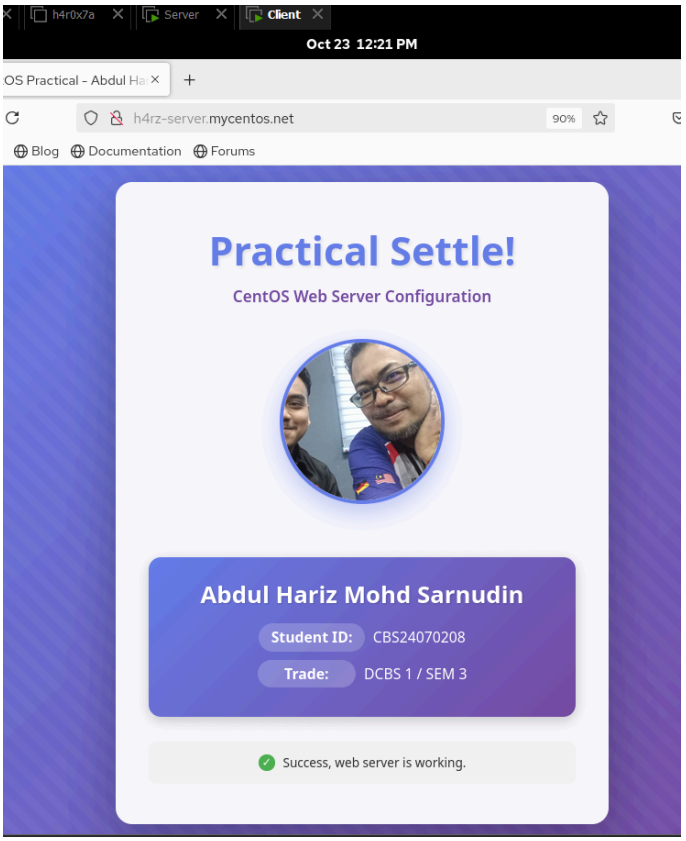
```
root@h4rz-client:~# nmcli connection down 'ens33'
Connection 'ens33' successfully deactivated (D-Bus active path: /org
root@h4rz-client:~# nmcli connection up 'ens33'
Connection successfully activated (D-Bus active path: /org/freedeskt
root@h4rz-client:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel
    link/ether 00:0c:29:5f:68:b9 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c295f68b9
    inet 172.30.1.213/24 brd 172.30.1.255 scope global dynamic nopre
       valid_lft 7199sec preferred_lft 7199sec
    inet6 fe80::20c:29ff:fe5f:68b9/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
root@h4rz-client:~#
```


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Troubleshooting	1) Problem : Name resolution doesn't work on client.
	
	`client@h4rz-client:~$ nslookup h4rz-server.mycentos.net`
	`;; communications error to 172.30.1.110#53: timed out`
	`;; communications error to 172.30.1.110#53: timed out`
	`;; Got SERVFAIL reply from 172.30.1.110`
	`Server: 172.30.1.110`
	`Address: 172.30.1.110#53`
	
	`** server can't find h4rz-server.mycentos.net: SERVFAIL`
	
	`client@h4rz-client:~$`
	
	Solution : I recheck the configuration file (/etc/named.conf) again and notice that my domain zone is wrong. Instead of mycentos.net, I put centos.net :
	
	`// Forward Zone - mycentos.net`
	`zone "centos.net" IN {`
	`type master;`
	`file "mycentos.net.zone";`
	`allow-update { none; };`
	`};`
	
	2) Problem : Cannot use and install dhcp (CentOS 10 doesn't support dhcp)
	
	Solution : Install service "kea-dhcp4" as an alternative to dhcp.
	
	`root@h4rz-server:~# dnf install -y kea`
	`Last metadata expiration check: 3:02:02 ago on Thu 23 Oct 2025 09:19:58 AM +08.`
	`Package kea-3.0.0-2.el10.x86_64 is already installed.`
	`Dependencies resolved.`
	`Nothing to do.`
	`Complete!`
	
	3) Problem : Client cannot access server's web page.
	
	Solution : Setup and allow firewall for service httpd, "firewall-cmd --add-service=http --permanent" and reload "firewall-cmd --reload".
		10

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Can now access server's webpage :



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Knowledge	DNS - Domain Name Services in analogical terms is like a “note book” or a dictionary for computers. It resolves a domain name to an ip address, for example gmi.edu.my -> 192.168.100.100. So reverse DNS is just vice versa where IP address resolves to domain, 192.168.100.100 -> gmi.edu.my. Simply said, DNS = Computer’s Note Book to resolve ip address and domain name. DHCP - Dynamic Host Configuration Protocol is a network protocol in assigning an ip address automatically. Analogically, DHCP is like a ticket clerk at a cinema. It gives a different ticket to customers every time they come and paid for the movie. Same goes to DHCP, every time a device joined the same network as the DHCP Server. It will automatically lease an IP Address to each device connected. Leased ip addresses range can be setup in the configuration file. Web Server - A web server is where a web site or web page will be hosted on. Web server typically uses HTML for their web page. Website has an IP addresses, but you can assign domain to it. Like we did in our practical, where we setup a web server. Supposedly we can open the web page by typing http://172.30.1.100, but since we configured a virtual host and domain for the website. We can open it by typing http://h4rz-server.mycentos.net. This is also way safer in terms of web security, instead of displaying the bare IP address, we uses a domain name to hide the IP.	**5**

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